VENKATA GANJI

AI INFRASTRUCTURE & PLATFORM ENGINEER

GPU MODEL SERVING | KUBERNETES | SEARCH & VECTOR INFRASTRUCTURE

Concord, CA | 925-272-9870 | venkataganji.careers@gmail.com | No sponsorship required

linkedin.com/in/venkata-sai-ganji-3a30a2a4 | Architecture Portfolio https://offroadguy.github.io/

AI platform engineer with 12 years building and operating production distributed systems.

Co-owned Apple's internal enterprise e2e private AI serving platform: GPU infrastructure, model onboarding, ingestion, vector search, and production reliability. Combines architecture with hands-on automation, performance engineering, and incident response.

AI PLATFORM OWNERSHIP

GPU model serving Owned model-weight serving and onboarding tooling; rapid prototyping and decommissioning.	**Search & vector databases** Owned Solr; co-owned Elasticsearch and Milvus: provisioning, performance tuning, tooling and IaC.
Ingestion & inference platforms Co-owned Kubernetes platform engineering and operations; GitOps onboarding with developers.	**GPU Kubernetes operations** Co-owned GPU cluster setup, upgrades, patching and production operations.

PLATFORM SCOPE Enterprise chat | Document intelligence | Agent applications | Studio UIs & APIs

Platform footprint: AWS | GCP | Alibaba Cloud | Apple private cloud

15,000+ collections Project embeddings | 5 Solr clusters	**~7.0s to ~1.9s** Vector-query latency	**$500K+ combined savings** Solr + Mac infrastructure

PROFESSIONAL EXPERIENCE

Apple

Apr 2024 - Aug 2026

AI Platform Engineering / infrastructure / Operations

Contract via Mphasis Corp | Enterprise AI infrastructure and platform operations

NVIDIA AWS Kubernetes Terraform Python Helm Argo CD
vLLM Solr Elasticsearch Milvus Prometheus Grafana

- Co-owned platform engineering and Kubernetes/EKS GPU lifecycle on AWS EC2 (NVIDIA V100, A10G, L40S, A100): setup, upgrades, patching, and operations. Application-cluster configurations specified 226 workers across two AWS regions.
- Owned model-weight onboarding and serving tooling using Helm/Argo CD and vLLM: up to eight-way tensor parallelism, GPU memory sizing, context limits, and LoRA; built S3-to-FSx weight staging with checksum validation and read-only mounts.
- Developed Go tools for model-weight checksums and Kubernetes Kafka topic/user resource generation.
- Partnered with developers on ingestion/inference GitOps onboarding across Dev/UAT/Prod; West-region document-processing configurations specified 105 parsing, chunking, and PDF-conversion replicas.
- Owned Solr 9.9 infrastructure, performance engineering, IaC, and lifecycle tooling at ~200-node, multi-region scale; supported 15,000+ project-embedding collections across five clusters in one region, with at least 90% active. Co-owned Elasticsearch and Milvus.
- Reduced vector-query latency from ~7.0s to ~1.9s through JVM/GC, OS, cache, query, and index tuning. Migrated 200 Solr nodes to Graviton/ARM64, saving 19.8% ($266K+ annualized); built provisioning, backup, and recovery tooling.
- Co-owned logging, dashboards, alerting, and incident response; investigated failures using network PCAP, p95/p99 latency, request/ingestion queue backlogs, and Kubernetes PVC diagnostics.
- Owned Mac conversion infrastructure, IaC, and tooling for Keynote, Numbers, and Pages to Office/PDF; productionized EC2 Mac conversion with developers, saving ~$16K/month.

MODEL PLATFORM Apple foundation models; Qwen3.5-397B-A17B-FP8; Qwen3-32B; Mixtral-8x22B-Instruct-v0.1; gpt-oss-120b; Pixtral-12B-2409.

CLOUDWICK TECHNOLOGIES INC | APR 2014 - APR 2024

Amgen
Sep 2023 - Apr 2024

AWS Cloud Solutions & Migration Engineer

Cloudwick client engagement

- Migrated **100+ AWS accounts** to Control Tower and Organizations using repeatable Python, CloudFormation, CDK, and SSM workflows; implemented secure S3 migrations with IAM/KMS.
- Tuned vector databases, resolved GenAI ingestion failures, and built Kubernetes deployment pipelines and Python monitoring integrations.

Freddie Mac
Jan 2019 - Aug 2023

AWS Cloud Solutions Architect & Infrastructure Consultant

Cloudwick client | Technical leadership for 3-8 engineers across 5+ projects

AWS • Disaster recovery S3 • PingFederate K8s Solr HBase Databricks

ARCHITECTURE	SOFTWARE ENGINEERING	PRODUCTION OPERATIONS
Rearchitecture & cloud migration	Spark, Python & ingestion pipelines	IaC, performance & disaster recovery

- **Architecture and migration.** Led three mission-critical migrations across distributed data and live OLTP/web platforms; rearchitected mortgage-document systems from Hadoop/Solr to Elasticsearch/OpenSearch and S3.
- **Software engineering.** Led Spark migration-job development for Solr-to-Elasticsearch and HBase-to-S3; designed Databricks, AWS Glue, Talend, and S3 ingestion frameworks and built Python/Airflow automation.
- **Infrastructure and operations.** Built Terraform, CloudFormation, and Ansible provisioning; automated snapshots and bootstrapping, supported performance testing, and resolved platform and release failures.
- **Disaster recovery.** Delivered a production hybrid-cloud DR environment within six months with ~15-minute RPO and ~2-hour RTO using cross-region replication, S3, Airflow, Python, and Elasticsearch.

Cyber Data Lake / Amorphic
Nov 2016 - Dec 2018

Lead Solutions Engineer

• PCAP • NetFlow • Telemetry Rust exposure Bash Python

Supermicro • SolrCapture • IXIA CentOS Ansible

OWNED INFRA & HARDWARE	OWNED NETWORK & CAPTURE	OWNED TOOLING & PERFORMANCE
Supermicro servers, high-throughput NICs	IXIA, ToR / supervisor switches, packet capture	Solr analytics, monitoring & performance tooling

- Owned the three-rack cyber data lake infrastructure end to end: hardware/network integration, operational and performance tooling, and tuning of 40 Gb/s packet-capture technology.
- Built correlated ingestion of PCAP, NetFlow, application logs, and firewall events across Hadoop, Kafka, Spark, Solr, HBase, and Elasticsearch; built Prometheus monitoring for capture agents.

EARLIER DISTRIBUTED SYSTEMS DELIVERY

AWS Azure exposure Hadoop Databricks Python Bash Ansible

Selected clients | Senior Operations Solutions Engineer | Feb-Oct 2016
Daimler, SimplyMeasured, Classmates / The Battle Group: Hadoop builds, Ansible automation, security, benchmarking, and production readiness.

Amobee / Turn | Operations Solutions Engineer | Apr 2015-Jan 2016
Operated Hadoop/HBase at up to ~400 nodes per cluster and 30 PB total capacity; delivered HA/DR, fleet refreshes, and Python/JMX monitoring.

Limelight Networks | Operations Engineer | Apr 2014-Apr 2015
Designed HA/DR for Hadoop billing, maintained ~80-node video analytics, and tuned Hive/MapReduce.

EDUCATION

Master's in Computer Science, Texas A&M University, 2014.

Additional graduate study: Montana State University, 15 credits. Bachelor of Technology, 2011.